Project 2 Documentation

Project: G12_Project2_2

Scope: scheduler-focused xv6 tree with an active lottery scheduler and two included reference/demo sources for MLFQ and RMS concepts.

1. Overview

In the current integrated-kernel tree, the active kernel scheduler remains the lottery scheduler implemented in kernel/proc.c. The project also includes two reference files from other branches: mlfq.c from 24je0651 and user/rmstest.c from Yashwanth.

These reference files are included for documentation and experimentation. They do not replace or modify the active kernel scheduler in this tree.

2. Active kernel implementation

- Scheduler type: lottery scheduling
- Main implementation file: kernel/proc.c
- Per-process scheduler state: tickets and sched_count in kernel/proc.h
- Default ticket value: 10 per process during allocation
- Fork behavior: child processes inherit parent ticket count
- Selection path: scheduler sums runnable tickets, draws a winning ticket, and runs the matching process

2.1 Lottery scheduler logic

The scheduler scans all RUNNABLE processes, normalizes the effective ticket count, computes cumulative ticket ranges, and uses a pseudo-random draw to select the winning range. The selected process then runs until it yields or blocks. sched_count records how many times a process is chosen.

3. Reference scheduling sources

3.1 MLFQ reference source

mlfq.c is a standalone mock simulation sourced from branch 24je0651. It models a 3-level multi-level feedback queue with time quanta of 1, 2, and 4 ticks.

- Tracked state: arrival time, burst time, remaining time, completion time, turnaround time, waiting time, queue level, queue ticks
- Behavior: always search higher queues first, execute one unit of work, and demote a process when its current queue quantum is exhausted
- Purpose: compact reference for queue movement and accounting without kernel integration

3.2 RMS demo source

user/rmstest.c is a reference demo sourced from branch Yashwanth. It models three periodic tasks with different periods and derived priorities to illustrate a rate-monotonic style setup.

- T1: period 4, derived priority 250, burst 2
- T2: period 6, derived priority 166, burst 3
- T3: period 12, derived priority 83, burst 4
- Purpose: source-level RMS demonstration only; not wired into the active scheduler in this tree

4. Features

- Active lottery scheduler in the kernel

- Per-process ticket accounting and schedule counters
- Included MLFQ reference source for alternative-policy study
- Included RMS demo source for alternative-policy study
- Clean separation between active scheduler code and reference/demo material

5. Testing and validation

Testing for Project 2 is split between the active kernel scheduler and the included reference sources.

- Active scheduler: build and run xv6 under CPU contention to observe lottery selection behavior
- MLFQ reference file: compile and run as a host-side C program if a standalone simulation is needed
- RMS reference file: review or adapt the source if RMS-specific task behavior needs to be demonstrated in a separate integration branch

The current tree does not switch the kernel scheduler to MLFQ or RMS. Those models remain documentation and reference material unless additional integration work is performed.

6. Build and execution

```
cd G12_Project2_2
make qemu
```

This builds and runs the active lottery-scheduler xv6 tree.

7. File map

- kernel/proc.c: active lottery scheduler implementation
- kernel/proc.h: per-process scheduler state fields
- mlfq.c: standalone MLFQ mock simulation from 24je0651
- user/rmstest.c: RMS demo source from Yashwanth
- README.md: short project summary for the current integrated-kernel state

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